

Grayson Clark

Portfolio: faultypine.github.io

Github: github.com/FaultyPine

Education: Indiana University - BS in Computer Science August 2020 - July 2024

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EXPERIENCE

- Blizzard Entertainment - Diablo** In-Person - Full time
Engine Programmer May 2024 - Present
Rewrote our asset compilation/processing pipeline to use a proprietary technology built by Activision for COD - yielded speedups for downloading & loading asset content. Transitioned our internal scripting runtime from Lua 5.1 to [Luau](#). Transitioned multiple systems use the [ECS](#) paradigm, and contributed to ECS-related infrastructure (multithreaded systems graph, runtime reflection/game data binding). Wrote a "walla" system for audio. Owned console (xbox, playstation) efforts to update and manage SDKs. Various on-console debugging to improve memory usage. Rewrote core input systems to support future console porting efforts.
- Blizzard Entertainment - Diablo 4** In-Person - Summer Internship
Engine Programmer May 2023 - August 2023
Worked closely with designers, artists, QA, and programmers to reduce engine/editor friction and improve iteration time. Wrote docs for various parts of the codebase, including for a proprietary Blizzard library. Integrated the C/C++ Hot-Reload tool Live++ into the Diablo proprietary engine. Reduced average programmer iteration cycle from avg. 5 minutes for compile-link-reboot to <10 seconds.
- ThirdPixel Interactive - Smack Studio** Remote - Contractual
Netcode Consultant February 2022 - March 2022
Analyzed the team's codebase to provide advice about recommended project architecture to facilitate their netcode solution
- Indiana University Bloomington** In-Person
Undergraduate Instructor - Introduction To Computers and Programming August 2022 - March 2024
- Deerfield Park District** In-Person - Seasonal
Elementary School Summer Camp Counselor May 2015 - July 2021

PROJECTS

- Mindseye Game Engine:** (WIP) Hobby engine to practice engine programming concepts. Currently features an asset and serialization system built on top of a robust clang-parser-based reflection system. Bonus feature: [a compile-time arbitrary code execution engine](#), similar in spirit to Zig's 'comptime' concept
- Brawlback - multiplayer client (Deterministic Lockstep, Speculative Execution, Client Synchronization, Peer-To-Peer Networking):** Peer-To-Peer Networking client for Super Smash Bros Brawl with Integrated Matchmaking that allows players to play with each other even in volatile or poor network conditions. Written in C++ and PowerPC Assembly, uses the [Rollback Netcode](#) technique
- Tiny Engine:** Created a lightweight framework of C++ APIs wrapping OpenGL and GLFW for videogames programming. Features include hardware-accelerated rendering, debug shape drawing, spritesheet parsing/animation, 3D mesh (.obj) loading/rendering, shader API, postprocessing, and a simple job (multithreading) system.
- Incremental Rollback :** A novel approach to state rewinding in emulator-based games. Utilizes dirty page tracking to save memory differences from one game frame to the next. Allows for rewinding an emulator's gamestate (opaque binary blob) efficiently - only dealing with parts of relevant memory that actually changed frame-to-frame.
- HewDraw Remix - overhaul modification (Rust, Code Injection):** (Formerly lead developer) open source modification of Super Smash Bros Ultimate. Major contributions included a full rewrite of the codebase from C++ to Rust in a 2 month period, a Continuous Deployment system, and a user-facing launcher/updater application. At time of writing we have around 25 contributors and 4000 active players. The project is composed of approximately 100,000 lines of Rust, as well as python scripts for automating various tasks and infrastructure
- UltimateModShop - User-facing shop/management app (GUI, Web API, Embedded Software):** a homebrew application written in C++ for the Nintendo Switch that allows users to download, install, and manage Smash Ultimate mods directly on their switch, using the GameBanana web API.
- Reverse Engineering / Skyline Plugins (Embedded Software, Reverse Engineering, Code Injection):** Utilized deep knowledge of C++/Assembly and an intuition for common code practices to Reverse Engineer parts of the Super Smash Bros Brawl and Super Smash Bros Ultimate executables. Interfaced with the Nintendo Switch's developer SDK to perform file IO, networking, etc. Used Reverse Engineered code and the [Skyline library](#) to properly perform code injections and overwrite game behavior with our desired behavior. Written in Rust. Created some [small projects](#) and was previously lead developer on [HewDraw Remix](#)
- Wave Function Collapse Algorithm Visualizer (Algorithms, Data Structures):** Implemented the Wave Function Collapse algorithm in JavaScript
- Unity Projects:** Created a physically [accurate solar system simulation with procedurally generated planets](#). Created various Unity projects that recreate game mechanics from popular games. Includes Cel Shading, Particle Systems, UI Integration, Image Effect Shaders, Water Shaders